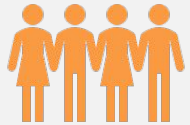


Information Systems

Year 9 Computing Technology

Learning Intention



Describe the purpose of connecting people and systems



Differentiate between data and information

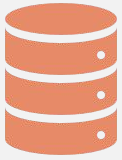
Success Criteria



I can describe the purpose of an information system and a digital system.



I can explain the difference between data and information.



I can outline the components and categories of an information system.

What are Information Systems?



A system is a group of connected components working together to achieve a goal



Examples of systems include biological systems (respiratory system) and natural systems (solar system)



A **digital system** combines hardware and software to perform tasks



When a digital system also includes **data, people and processes**, it becomes an **information system**

Digital Systems

- Consist of hardware, software and often networks
- Can operate across large distances using the internet
- Enable activities such as video calls, online banking and smart home control
- Form the technical foundation of information systems

Purpose of an Information System



To collect, process and transform data into useful information



Used to solve problems or meet identified needs

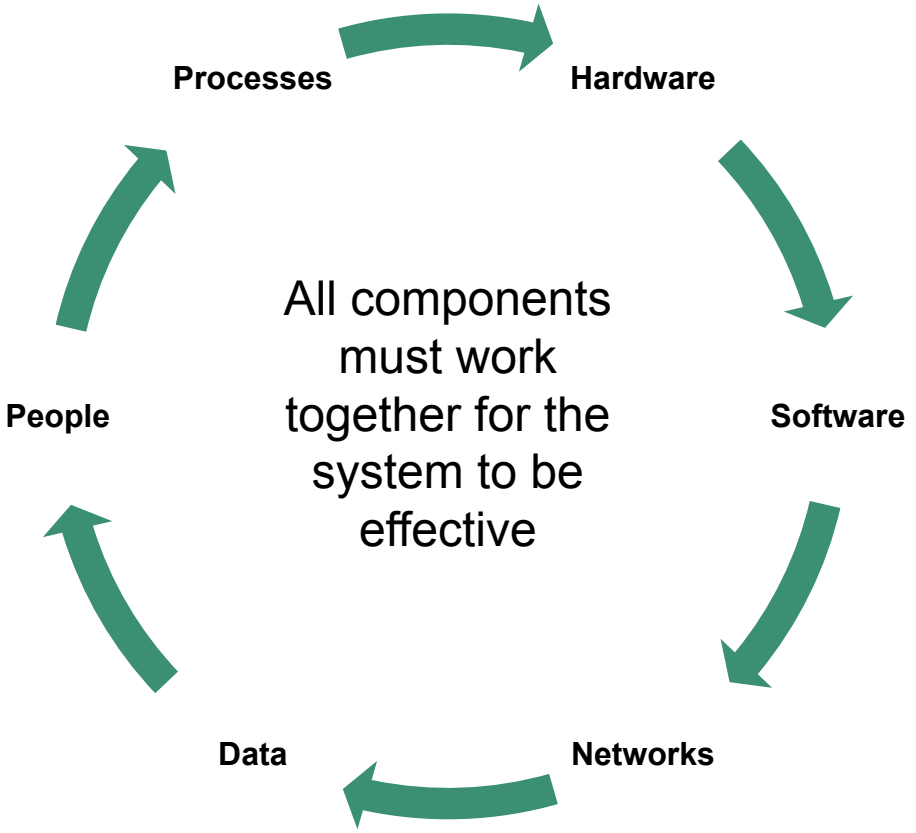


Supports decision-making, automation and efficiency



Commonly used in organisations and enterprises

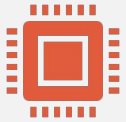
Components of an Information System



Hardware



DEFINITION: Physical components of an information system



Includes computers, servers and mobile devices



Uses input and output devices such as keyboards, cameras, printers and scanners



Requires networking hardware like routers, cables and access points

Software

Instructions that tell hardware how to process data



Written in programming languages and converted to machine code

Includes application software (e.g. browsers, apps, databases) and system software (operating systems and utilities)

Enables data processing, storage and user interaction

Networks



Connect two or more computers to share data



Can use wired connections (ethernet, fibre) or wireless connections (Wi-Fi, Bluetooth)



The internet is the largest global network

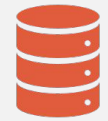


Enables real-time communication and remote access to systems

Data and information



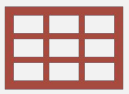
Data: raw facts collected, stored and processed by a system



Information: processed data that has meaning for users



Data is collected, stored, processed and displayed



Processing turns data into useful outputs for decision-making

People and Processes



People include users, operators and system maintainers



Human roles are still critical despite advances in AI



Processes are the rules and steps that guide system operation



Processes ensure consistency, accuracy and reliability

Structured Data



Quantitative data stored in predefined formats or models





Organised into records and fields within databases



Easier to store, search and analyse



Examples include names, dates, IDs and timestamps

Unstructured Data



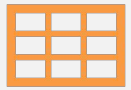
Does not fit predefined formats or models

Often complex and harder to analyse

Examples include photos, videos and audio

Requires technologies such as OCR and machine learning

The Three Vs of Data



Variety: different types of data (structured and unstructured)



Volume: the amount of data collected



Velocity: the speed at which data is generated



Used to determine appropriate data processing methods

Evolution of Data Analysis



Advances in AI allow analysis of unstructured data



Machine learning finds patterns without predefined models

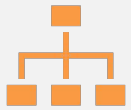


Enables connections between previously separate data sources



Supports deeper insights and automated decision-making

Categories of Information Systems



Management Information Systems (MIS)
Help organisations manage data and make decisions



Transaction Processing Systems (TPS)
Process everyday transactions like purchases and payments



Decision Support Systems (DSS)
Help people make complex decisions using data



Expert Systems
Act like a human expert by using rules and knowledge

Categories of Information Systems

Geographic Information Systems (GIS)
Use maps and location data to show patterns and trends

Enterprise Information Systems
Large systems that combine many functions across an organisation

Learning Management Systems (LMS)
Deliver and manage learning and track student progress

Multimedia Systems
Use text, images, audio and video to create interactive experiences

Databases and Data Warehouses



Databases store organised data



Data warehouses combine multiple databases

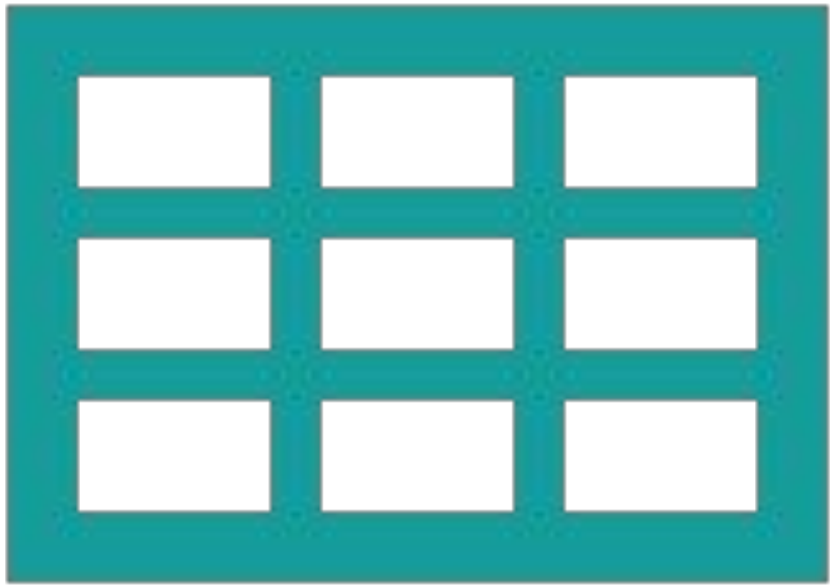


Enable data mining and advanced analysis



Can raise ethical concerns when combining personal data

Spreadsheet Software



- Used for quantitative data analysis
- Can support budgeting, tracking and modelling
- Allows custom solutions for small-scale systems
- Examples include Microsoft Excel and Google Sheets

Artificial Neural Networks



Programs that learn from experience



Improve performance through machine learning



Used for automated insights and decision support



Common in chatbots, recommendation systems and analytics